



Designing and Testing a Digital Front-End Block Using Verilog HDL

The Final Submission

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References

- 1- Polyphase FIR Filters
- 2- Polyphase Decomposition
- 3- Fixed-Point Arithmetic
- 4- IIR filter architecture
- 5- Cascaded Integrator-Comb (CIC) Filters

Abbreviation

Abbreviation	Description
IO	Input-Output
FD	Fractional Decimator
CIC	Cascaded Integrator-Comb
IIR	Infinite Impulse Response
FIR	Finite Impulse Response
LPF	Low Pass Filter
f_{o1}	Interference at 2 MHz
f_{o2}	Interference at 2.4 MHz
SNR	Signal to Noise Ratio
DUT	Design Under Test
AMBA	Advanced Microcontroller Bus Architecture
APB	Advanced Peripheral Bus

Top Level DFE Overview:

In modern digital communication and signal processing systems, the Digital Front-End (DFE) is responsible for preparing signals for further baseband processing. It typically performs sampling rate conversion, interference suppression, and spectral shaping.

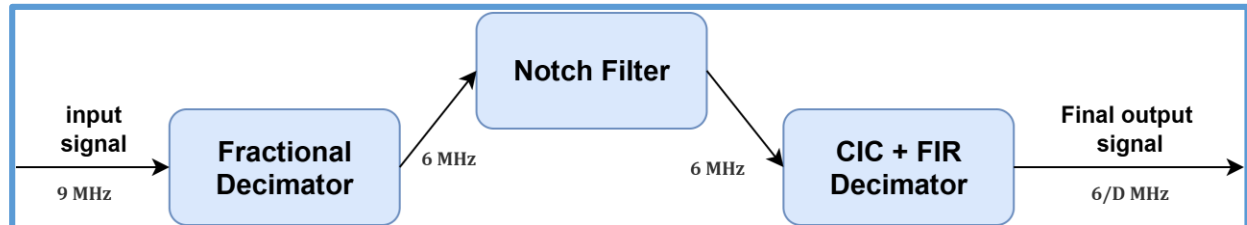


Figure 1: System block-level

The system receives the input signal with $F_s = 9$ MHz. The fractional decimator works on down sampling to $F_s = 6$ MHz, and then the notch filter suppresses unwanted narrowband interference. Finally, the CIC decimator reduces the sampling rate from $F_s = 6$ MHz to a much lower rate $F_s = \frac{6}{D}$ MHz.

System IO Table:

DFE IOs:

Table 1: Table of Digital Front End I/O

Signal name	Direction	Description
clk	Input	System clock
rst_n	Input	Asynchronous active-low reset
notch_enable	Input	Active-low bypassing for notch filter
cic_decimation_factor [4:0]	input	Average filter decimation factor
data_in [d_width-1 : 0]	input	Input data
data_out [d_width-1 : 0]	output	Output data
d_width equal to 16		

AMBA APB IOs:

Table 2: Table of APB I/O & DFE Signals

Signal name	Direction	Description
PCLK	Input	System clock
PRESETn	Input	Asynchronous active-low reset
PSEL	Input	Select. The APB bridge unit generates this signal to each peripheral bus slave.
PENABLE	input	Enable. This signal indicates the second and subsequent cycles of an APB transfer.
PWRITE	input	Direction. This signal indicates an APB write access when HIGH and an APB read access when LOW
PADDR [ADDR_WIDTH-1 : 0]	input	Address. This is the APB address bus. It can be up to 32 bits wide and is driven by the peripheral bus bridge unit.
PWDATA [DATA_WIDTH-1 : 0]	input	Write data. This bus is driven by the peripheral bus bridge unit during write cycles when PWRITE is HIGH.
PRDATA [DATA_WIDTH-1 : 0]	output	Read Data. The selected slave drives this bus during read cycles when PWRITE is LOW.
DFE Signals		
input_data [DFE_WIDTH-1 : 0]	input	Input data of the DFE
output_data [DFE_WIDTH-1 : 0]	output	Output data of the DFE

Stage 1 - Fractional Decimator:

Convert the input signal, from **9 MHz**, to **6 MHz** using fractional resampling. This is achieved using the L and M ($\frac{L}{M} = \frac{2}{3}$).

The Fractional Decimator Overview:

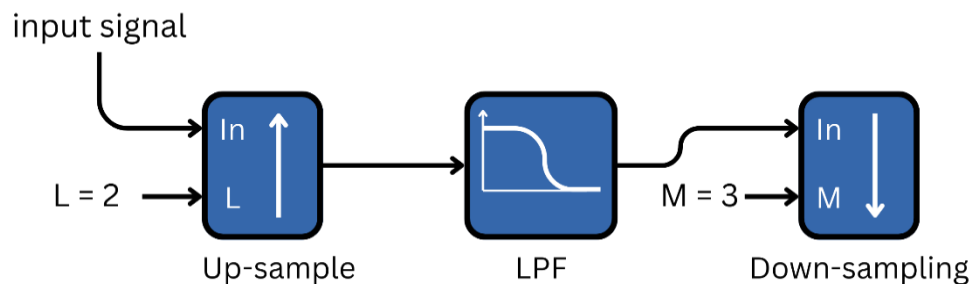


Figure 2: Fractional Decimator Operation

First, the input signal will be up-sampled by a factor of 2 ($F_s = 18$ MHz) and then removed of imaging due to up-sampling using FIR filter, and finally it is decimated by a factor of 3 to reduce the sampling rate ($F_s = 6$ MHz).

Upsampling by 2: increasing the sampling rate of a discrete-time signal, $x[n]$, by a factor of $L=2$, utilizing zero insertion.

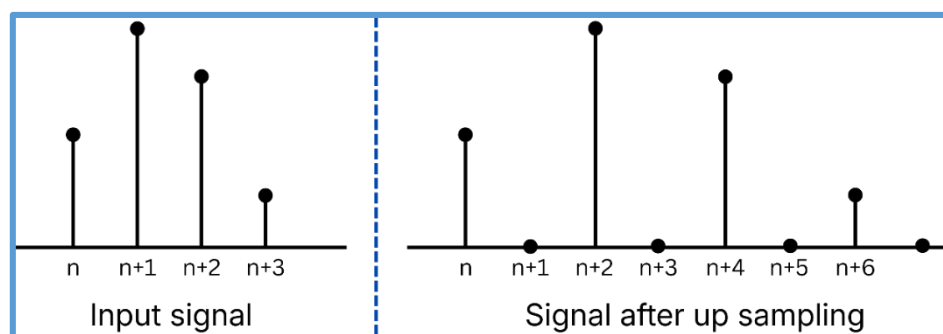


Figure 4 - Up sampling by factor of 2

Low pass FIR filter: is used before decimation for preventing aliasing and defining the new signal bandwidth, which is ≈ 3 MHz (new Nyquist).

Down sampling by 3: reduces the sampling rate by keeping one out of every 3 samples: $y[n] = x[3n]$.

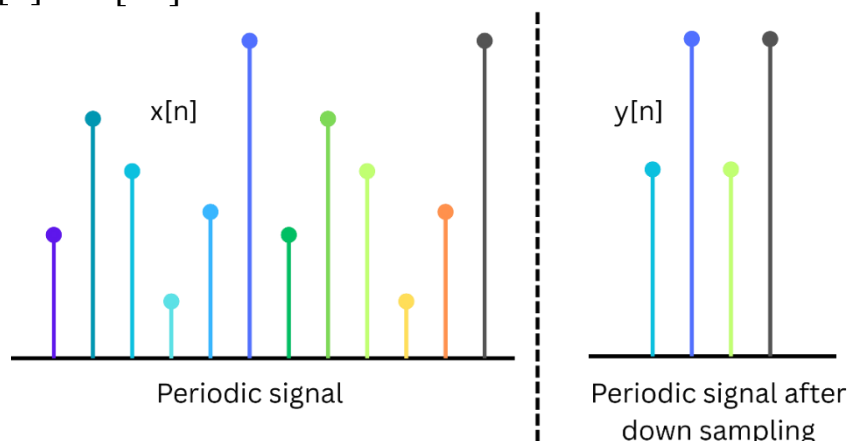


Figure 5 - Down sampling by factor of 3

1. Hardware Implementation:

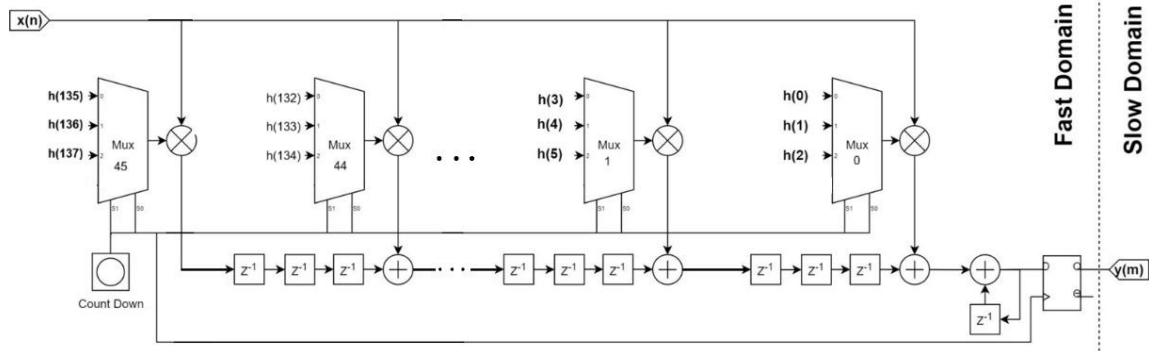


Figure 6 - Transposed Polyphase Decimator

The architecture is based on a Transposed Polyphase Filter.

For decimation by 3, we split the filter into 3 polyphase branches:

$$E_0[n] = h[3n]$$

$$E_1[n] = h[3n + 1]$$

$$E_2[n] = h[3n + 2]$$

MUX Blocks: each MUX selects one tap from its polyphase group.

2-bit Counter: selects the correct coefficient for each cycle.

Multiplier Blocks: each MUX output is multiplied by the current input sample $x(n)$.

Delay Chain: This delay structure stores partial FIR sums. In a transposed implementation, each register holds partial results for future outputs. The delay chain corresponds to tap delays, shared across cycles.

Adder Block: Each multiplier output is added to a stored partial sum. Over 3 cycles, each partial result accumulates all FIR contributions for one output sample. After the last cycle, the final output sample is complete.

Output Selector: The decimator outputs one sample every 3 cycles.

A modulo-3 counter enables output only when: cycle = 2 (end of polyphase accumulation). This sample is then labelled $y(m)$.

Simulation Diagrams:

Output Results:

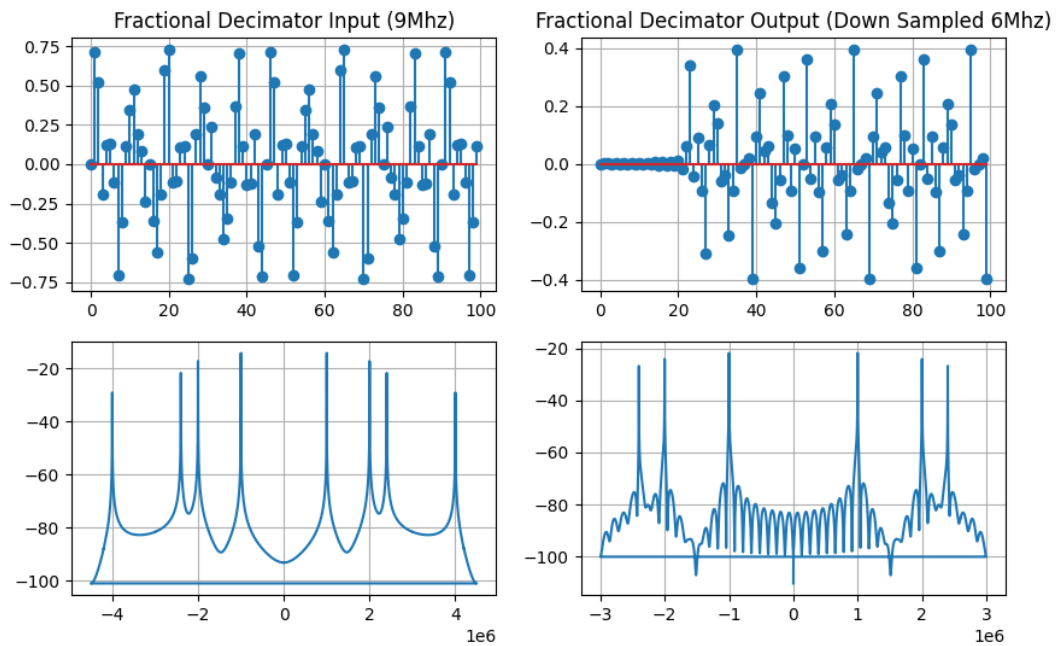


Figure 7 - Fractional Decimator Python model output

SNR: The fractional decimator shows moderate attenuation, which is expected due to normalized polyphase filter coefficients. The required accumulator width is small, indicating limited dynamic range growth. The SNR of 75.50 dB reflects excellent accuracy and very low quantization noise.

Stage 2 – Notch Filter:

Suppresses unwanted narrowband interferences at **5 MHz** and **2.4 MHz** relative to **6 MHz** (after the first stage – fractional decimator). It's resolved that the component of **5 MHz** will be folded at 2 MHz due to aliasing, **in case** of passing through the fractional decimator low pass filter.

Notch Filter Overview:

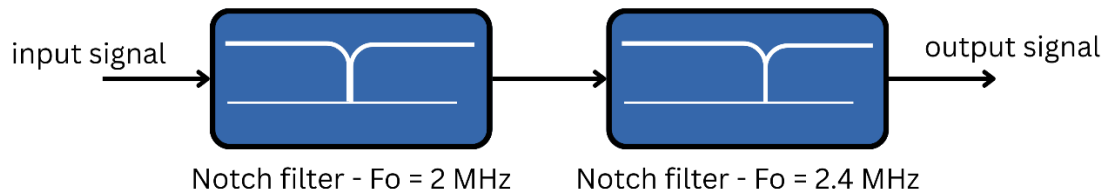


Figure 8 - IIR Notch Filter Block-Level

The input signal passes through two stages notch filter to remove the interference.

Filter's Equation:

$$H(z) = \frac{1 - 2r_z \cos(w_o) z^{-1} + r_z^2 z^{-2}}{1 - 2r_p \cos(w_o) z^{-1} + r_p^2 z^{-2}}$$

To achieve the BW < 50 KHz and notch depth > 50 dB, zero radius must be a little less than 1, and the pole radius too, to achieve stability.

Hardware Implementation:

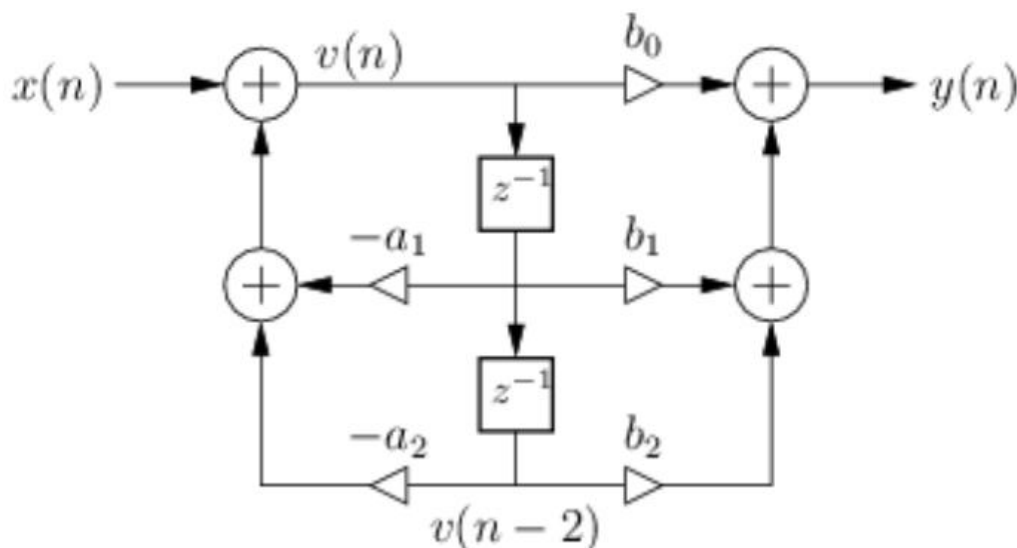


Figure 9 - Direct Form II implementation

The filter can be written as:

$$y[n] = B_0 x[n] + B_1 x[n-1] + B_2 x[n-2] - A_1 y[n-1] - A_2 y[n-2]$$

Notice that the system is LTI with memory, so values need to be stored in register through the computation.

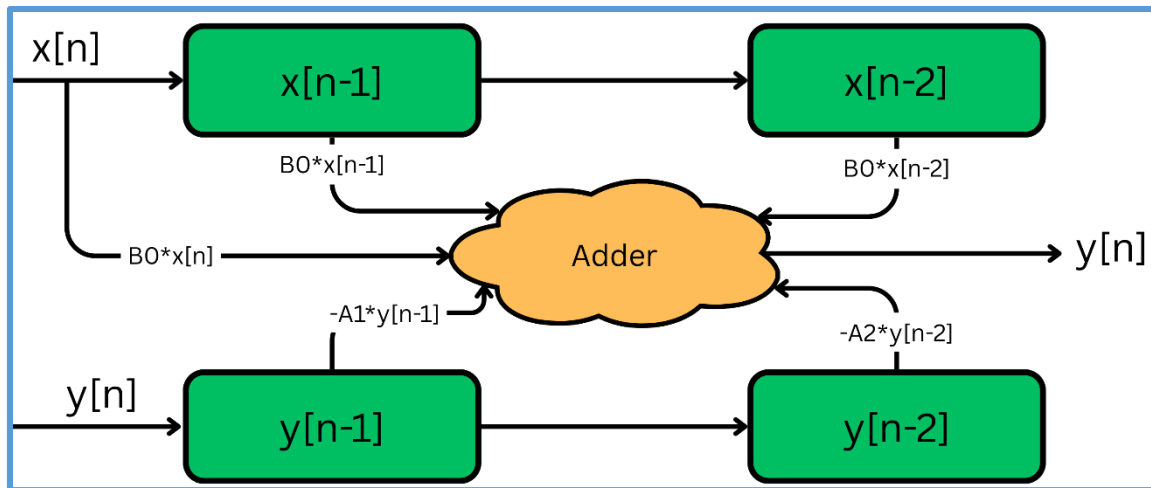


Figure 10 - One stage Notch filter using Hardware

Where B_0, B_1, B_2, A_1 , and A_2 are the filter coefficients, computed using the Python model using S16.14 fixed point format.

Sign	Integer	Fraction
1-bit	1-bit	14-bit

With range [-1.999: 1.999].

Coefficient	Floating point value	Fixed point value
$f_0 = 2 \text{ MHz}$		
B_0	1	16'h4000
B_1	-0.9999	16'h4000
B_2	0.9999	16'h4000
A_1	-0.970	16'h3E14
A_2	0.9409	16'h3C38
$f_0 = 2.4 \text{ MHz}$		
B_0	1	16'h4000
B_1	1.6180137633	16'h678E
B_2	0.999972	16'h4000
A_1	1.569492	16'h6473
A_2	0.9409	16'h3C38

1. The Frequency Response:

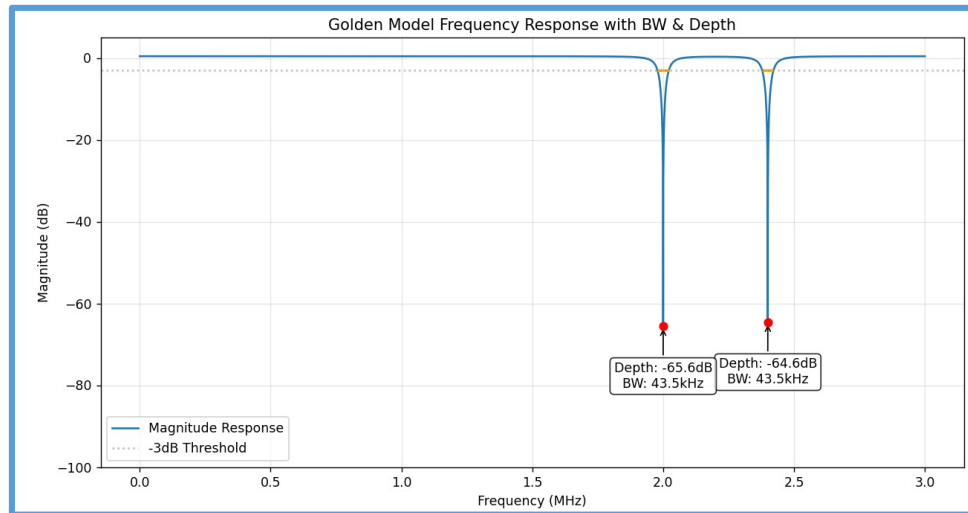


Figure 11 - Notch Filter Frequency Response

Simulation Diagrams:

The Output Result:

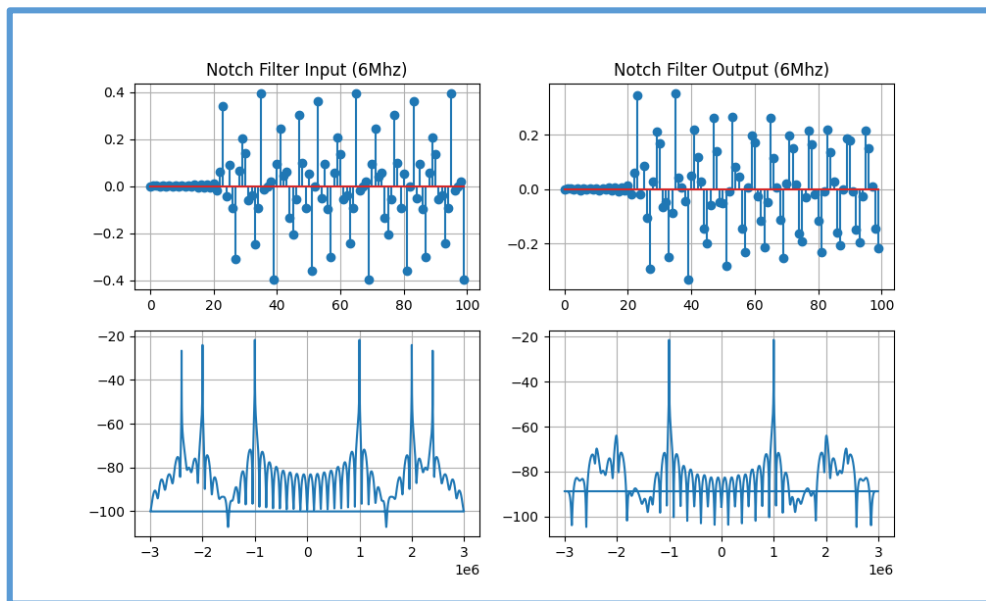


Figure 12 – Notch Filter Python Model Output

SNR: The IIR stage introduces additional attenuation, reducing signal amplitude further. An 18-bit accumulator is sufficient, but the SNR drops to **58.80 dB**, which is typical for fixed-point IIR filters due to feedback sensitivity and coefficient quantization.

Stage 3 – Cascaded Integrator-Comb:

The third stage of the Digital Front End (DFE) reduces the sample rate from 6 MHz to $\frac{6}{D}$ MHz using a Cascaded Integrator-Comb (CIC) decimator with a runtime-selectable decimation factor $D \in \{1, 2, 4, 8, 16\}$.

Functionally, the CIC stage:

- Produces 1 output for every D input, so the output sample is down sampled.
- Minimizes hardware cost (multiplier-free decimation), with fixed latency.

1. The CIC Filter Overview:

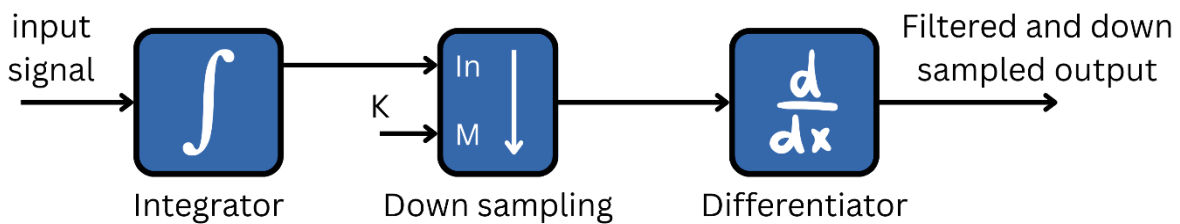


Figure 13 - CIC Block-level

The single-stage CIC first integrates the input by adding the current sampled input and the delayed sampled input, then decimates according to the selected value, and lastly, it differentiates the last two samples.

2. The Hardware Implementation:

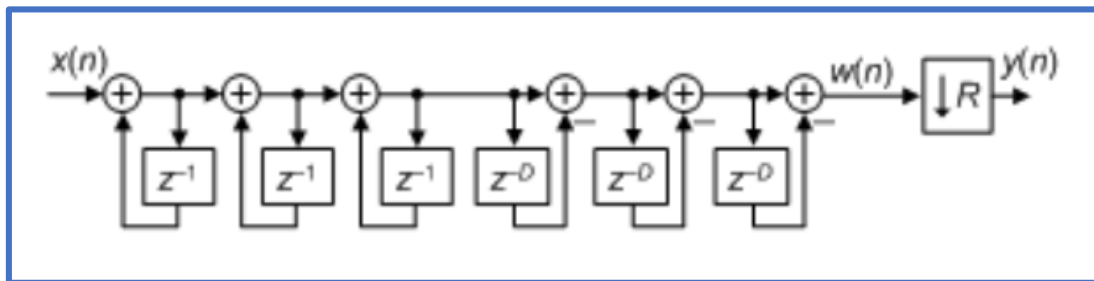


Figure 14 - CIC Architecture

Since we decimate the sampling frequency, aliasing must occur, but it is a trade-off between the number of stages and the droop that happens.

If we increase the number of stages, the filter becomes more anti-aliasing and image-rejection, but on the other side, the drooping happens that requires a compensation filter to equalize the drooping.

Thus, a sweet spot was chosen of 3 stages for the CIC filter, manipulating the position of the three integrators and differentiators, and leaving the down sampling at the end of the stage. This architecture is widely used.

Also, another thing to consider is that multiple stages give gain according to this equation:

$$Gain = D_{max}^N = 16^3 = 4096$$

Where D_{max} is the maximum decimated value, which is 16.

N is the number of cascaded stages of CIC.

Thus, sizing of the output should be done by accounting for the gain:

$$Bit_{growth} = \text{ceil}(N \cdot \log_2(16)) = 12 \text{ bits}$$

Then the internal width of our register should be:

$$Width = 16 + 12 = 28 \text{ bits}$$

Simulation Diagrams:

The Output Results:

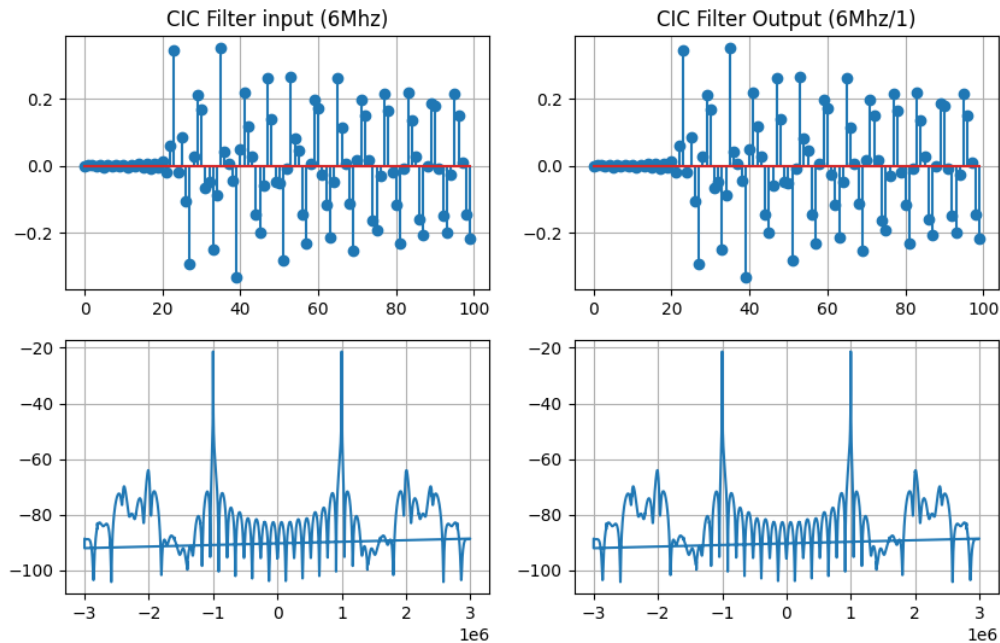


Figure 14 - Output for D = 1

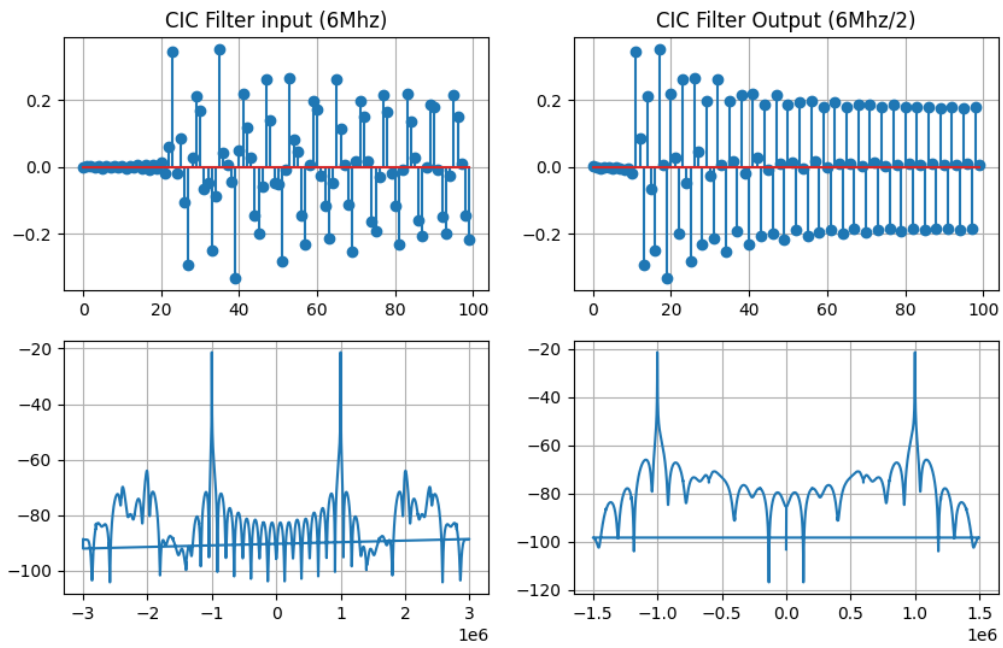


Figure 15 - Output for $D = 2$

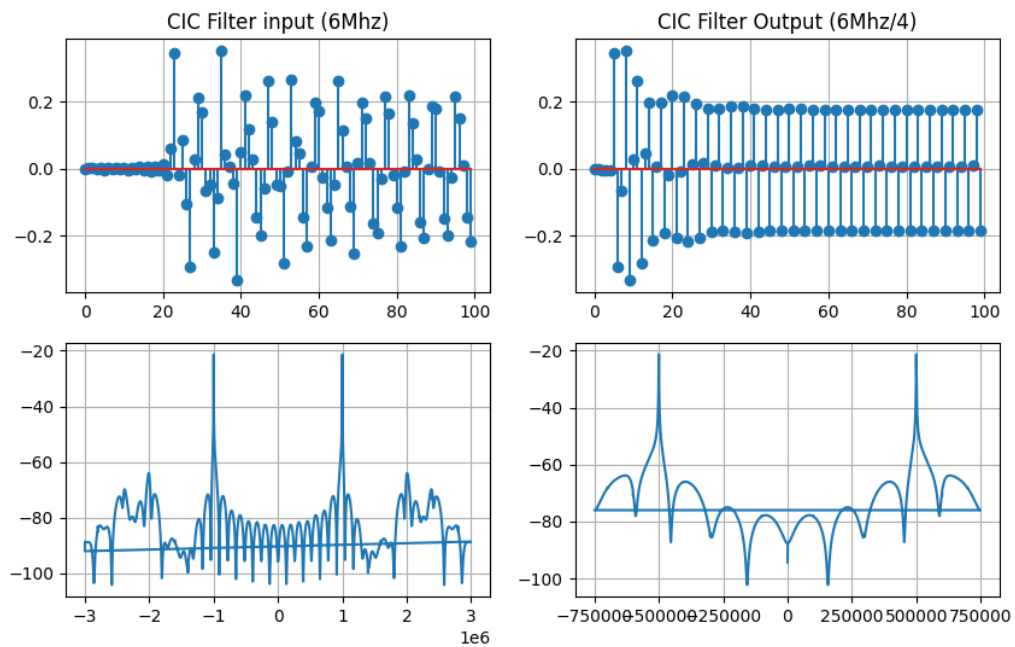


Figure 16 - Output for $D = 4$

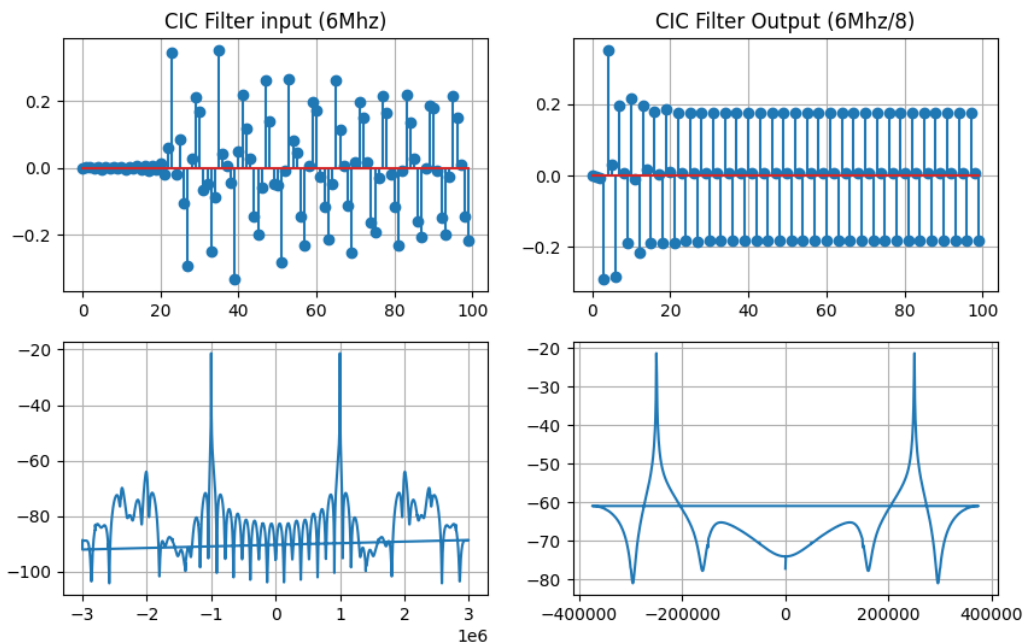


Figure 17 - Output for D = 8

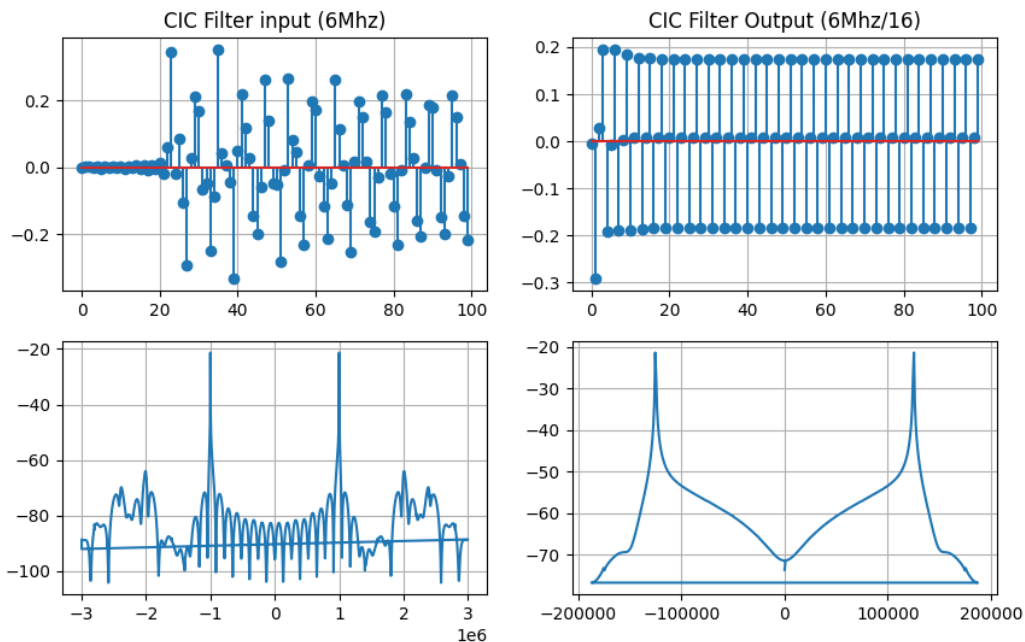


Figure 18 - Output for D = 16

SNR: The CIC filter exhibits a gain close to what is expected from its cascade structure. The **30-bit accumulator** is necessary to safely handle integrator growth without overflow. The **SNR of 59.41 dB** aligns with typical CIC behavior, where quantization noise and passband droop reduce accuracy compared to FIR-based stages.

The System Wrapper:

The system is wrapped with an APB interface to communicate with the microprocessor and adjusting the system control signals or coefficients.

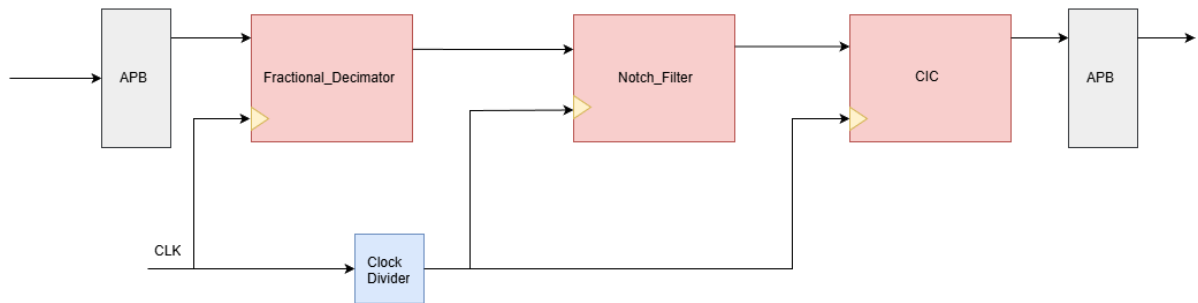


Figure 19 - The Whole System with APB Interface

Register File Content: eight registers with width size 32-bit.

Register	Content
R0	Control signals
R1	Status flags
R2 to R6	Notch filter coefficients

Registers Mapping:

R0	Enable					Bypass				CIC factor			[12:3]
	[0]	[1]	[2]	[3]	[4]	[5]	[6]	[7]	[8]	[9]	[10]	[11]	1'd0
	FD	F01	F02	CIC	clk_div	FD	F01	F02	CIC	Factor [2:0]		reserved	
R1	Reserved												

Register	[0:17]	[18:31]
R2	$B_{0-f_{01}}$	$B_{1-f_{01}}$
R3	$B_{2-f_{01}}$	$A_{1-f_{01}}$
R4	$A_{2-f_{01}}$	$B_{0-f_{02}}$
R5	$B_{1-f_{02}}$	$B_{2-f_{02}}$
R6	$A_{1-f_{02}}$	$A_{2-f_{02}}$

Enable: controls the block ON/OFF mode.

Bypass: skips the stage.

CIC factor: the decimation factor of the third block.

Fixed Point converting:

The notch filter coefficient is represented by S16.14 (due to the coefficient range), and the data output from the fractional decimator is represented by S16.15. Between the fractional decimator and the notch filter, there is a S16.15 to S16.14 converter.

Table of Accumulator Bits:

STAGE	Accumulator Bits
Fractional Decimator	18 bits
IIR filter	18 bits
CIC	28 bits

The regression test summary of the system

=====
REGRESSION TEST SUMMARY
=====
TEST 1: FRACTIONAL DECIMATOR

FD Status Num. of correct = 6712, Num. of errors = 0
TEST 2: NOTCH FILTER

Notch Status Num. of correct = 6712, Num. of errors = 0
TEST 3: CIC FILTER (Parameter Sweep)

CIC (D = 1) Status Num. of correct = 6709, Num. of errors = 0
CIC (D = 2) Status Num. of correct = 3354, Num. of errors = 0
CIC (D = 4) Status Num. of correct = 1677, Num. of errors = 0
CIC (D = 8) Status Num. of correct = 838, Num. of errors = 0
CIC (D = 16) Status Num. of correct = 419, Num. of errors = 0
TEST 4: SYSTEM LEVEL INTEGRATION

FD Status :: Num. of correct = 6711, Num. of errors = 0
Notch Status :: Num. of correct = 6711, Num. of errors = 0
CIC Status :: Num. of correct = 419, Num. of errors = 0

Figure 19 - The regression test summary

Waveforms output

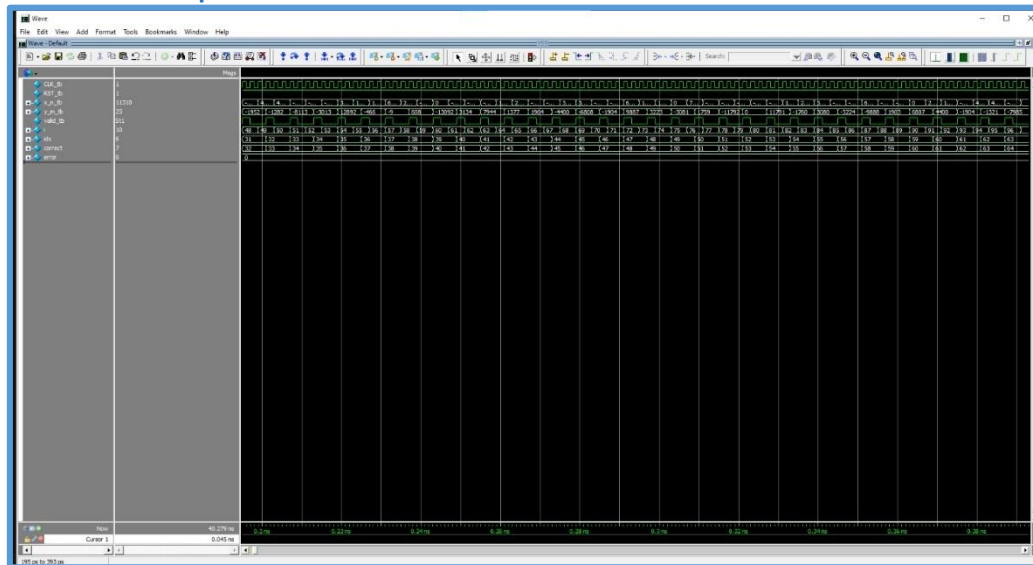


Figure 20 - Fracional Decimator Waveform



Figure 21 - IIR Waveform

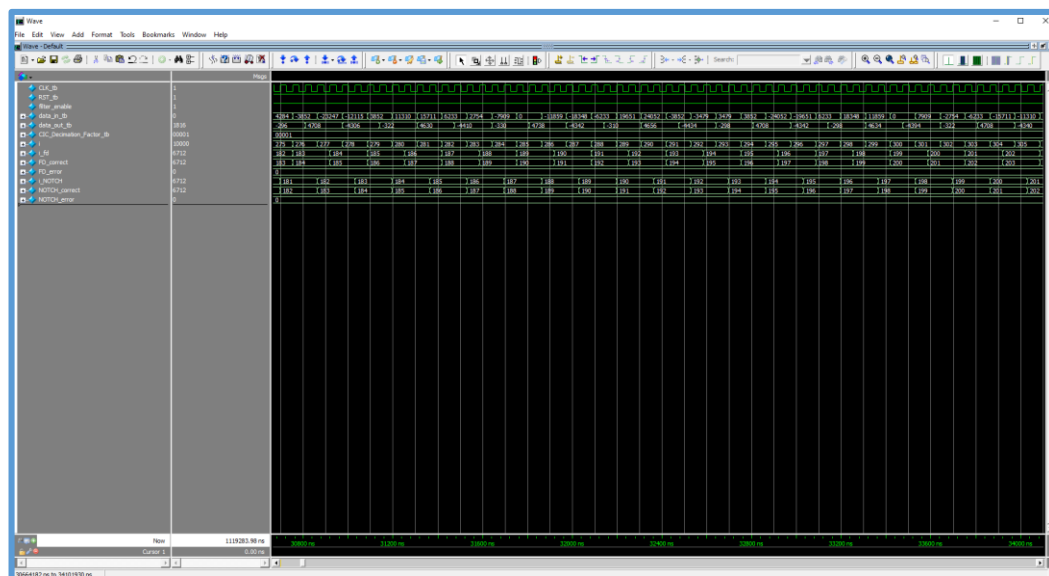


Figure 22 - CIC Waveform

FPGA Flow

1. Power consumption

As shown in the power report below the total chip power consumption is 0.253 Watt, and it also shows how much power did each block in the system consume.

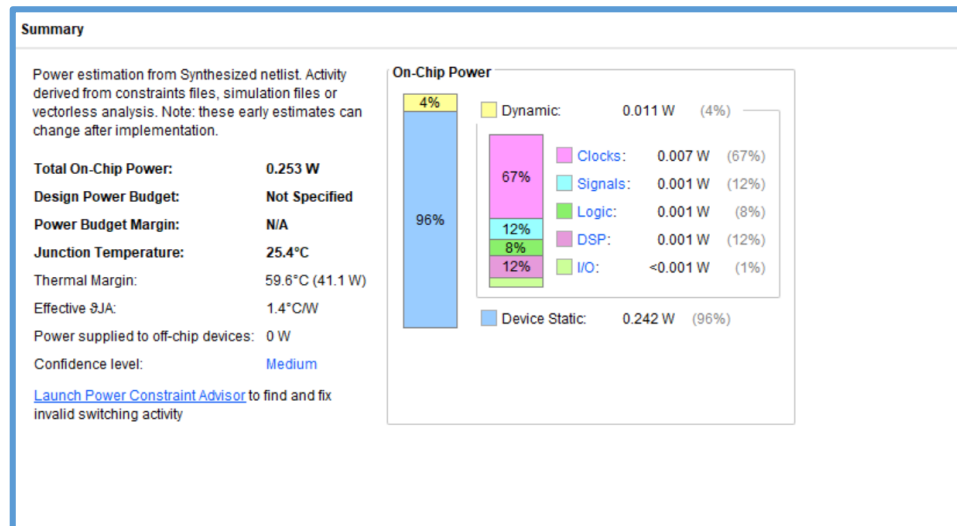


Figure 23 - Power Report by VIVADO

Utilization	Name	Clocks (W)	Signals (W)	Data (W)	Clock Enable (W)	Set/Reset (W)	Logic (W)	DSP (W)	I/O (W)
0.011 W (4% of total)	top_module								
0.01 W (4% of total)	u_dft_top (dfe_top)	0.007	0.001	0.001	<0.001	<0.001	0.001	0.001	<0.001
0.007 W (3% of total)	FR_D (Fractional...	0.006	0.001	0.001	<0.001	<0.001	<0.001	<0.001	<0.001
0.001 W (1% of total)	Notch_1 (Notch_...	<0.001	<0.001	<0.001	<0.001	<0.001	<0.001	0.001	<0.001
0.001 W (1% of total)	Notch_2 (Notch_...	<0.001	<0.001	<0.001	<0.001	<0.001	<0.001	<0.001	<0.001
0.001 W (<1% of total)	CIC_Filter (CIC)	<0.001	<0.001	<0.001	<0.001	<0.001	<0.001	<0.001	<0.001
<0.001 W (<1% of tot...	CLK_DIV (clk_div)	<0.001	<0.001	<0.001	<0.001	<0.001	<0.001	<0.001	<0.001
0.001 W (<1% of total)	Leaf Cells (518)								

Figure 24 - Detailed Each Block Consumption

2. Timing Summary

In this timing analysis it shown that we have setup time slack of 25.6 nano seconds meaning we can increase the operating period with the shown slack.

Design Timing Summary			
Setup	Hold	Pulse Width	
Worst Negative Slack (WNS): 26.654 ns	Worst Hold Slack (WHS): -3.303 ns	Worst Pulse Width Slack (WPWS): 27.428 ns	
Total Negative Slack (TNS): 0.000 ns	Total Hold Slack (THS): -1988.425 ns	Total Pulse Width Negative Slack (TPWS): 0.000 ns	
Number of Failing Endpoints: 0	Number of Failing Endpoints: 4294	Number of Failing Endpoints: 0	
Total Number of Endpoints: 10977	Total Number of Endpoints: 10977	Total Number of Endpoints: 3433	
Timing constraints are not met.			

Figure 25 - Timing Analysis by VIVADO

Noting that the hold time violation can be resolved in the implementation process and the reset is synchronized to the slowed clock.

3. Resource Utilization

For the area analysis the DFE block used a total of 59 DSPs, 3384 Slicers and 1288 LUTs.

Name	Slice LUTs (303600)	Slice Registers (607200)	DSP s (280 0)	Bonded IOB (600)	BUFGCTRL (32)
▼ N top_module	1288	3384	59	105	2
▼  u_dft_top (dfe_top)	1180	3113	59	0	1
 CIC_Filter (CIC)	110	87	0	0	0
 CLK_DIV (clk_div)	2	2	0	0	1
 FR_D (Fractional_Decimator)	919	2930	47	0	0
 Notch_1 (Notch_Filter)	75	46	6	0	0
 Notch_2 (Notch_Filter_0)	88	48	6	0	0

Figure 26 - Utilization Report by VIVADO

All files and results related to DFE system

https://drive.google.com/drive/folders/1AuHIYvV00HaUXZNyOFbr27-DkO1FXT_I?usp=drive_link